APPROVED OPERATIONAL RUNBOOK: FXT-RUN-003

Kafka Trade Event Pipeline Lag & Consumer Rebalance Failure Runbook

Application: Market Data & Trade Analytics	**Service:** Kafka-Broker-Cluster / Event-Processor	**Region:** AMER (New York Primary)
Environment: PROD	**Category:** Messaging & Real-time Streaming	**Version:** 1.8.3
Owner / Team: Event Infrastructure & Messaging APS	**Last Updated:** 2026-07-28	**Doc ID:** FXT-RUN-003

1. Operational Symptoms & Issue Summary

Issue Summary: Consumer lag on 'fxt.orders.executed.v1' surging past 200,000 messages, resulting in delayed market data feeds and stale risk calculation for trading desks.

Detectable Symptoms & Alerts:

- Alert: 'KAFKA_CONSUMER_LAG_HIGH: Group fxt-risk-calc-group lag > 250,000 on topic fxt.orders.executed.v1'
- Log Warning: 'CommitFailedException: Offset commit cannot be completed since the consumer is not part of group'
- Log Error: 'Heartbeat thread error: member has been kicked out of group due to max.poll.interval.ms timeout'
- Trader symptom: Pricing dashboard shows 45-second latency indicator flag

2. Root Causes & Technical Diagnosis

- Heavy JSON parsing or external database lookup inside consumer loop causing processing time to exceed max.poll.interval.ms (300,000ms).
- Frequent rebalance storms caused by an unstable node repeatedly joining and leaving the consumer group.
- Uneven partition skew where Partition 4 receives 70% of peak volume.

3. Mandatory Safety Directives (DO NOT DO)

CRITICAL OPERATIONAL WARNING: **DO NOT reset consumer group offsets to 'latest' without explicit approval from Risk Management; doing so will permanently drop execution records.**
CRITICAL OPERATIONAL WARNING: **DO NOT scale the consumer group pod count higher than the number of topic partitions (16 max partitions).**
CRITICAL OPERATIONAL WARNING: **DO NOT restart Kafka broker nodes during market volatility spikes (e.g. Non-Farm Payroll, FOMC release).**

4. Step-by-Step Resolution Procedures

Step 1: Inspect Consumer Group Lag and Partition Distribution

Execute Kafka consumer group description command to view current lag per partition.

Command (Bash/Shell):

```
kafka-consumer-groups.sh --bootstrap-server kafka-prod-01:9092 --describe --group
fxt-risk-calc-group
```

Step 2: Identify Failing Consumer Pods

Search Kubernetes pod logs for Rebalance and CommitFailed exceptions.

Command (PowerShell):

```
kubectl logs -l app=fxt-risk-calc -n fxt-prod --tail=100 | Select-String -Pattern
'CommitFailedException|rebalance'
```

Step 3: Scale Consumers Up to Match Partition Count

Scale up consumer deployment to 16 replicas (1 consumer per partition max).

Command (PowerShell):

```
kubectl scale deployment/fxt-risk-calc --replicas=16 -n fxt-prod
```

Step 4: Execute Dry-Run Offset Verification If Rewind Is Needed

Never execute offset reset without running --dry-run first.

Command (Bash/Shell):

```
kafka-consumer-groups.sh --bootstrap-server kafka-prod-01:9092 --group fxt-risk-calc-group
--reset-offsets --to-offset 8940201 --dry-run --topic fxt.orders.executed.v1
```

5. Post-Recovery Verification Procedures

V-1: Confirm 'TOTAL_LAG' metric is steadily decreasing at minimum 5,000 messages/sec.

V-2: Verify all 16 consumer members are assigned distinct partition IDs with state 'Stable'.

V-3: Check Risk Dashboard latency counter returns to < 200ms.

6. Dependency-Aware Service Recovery Sequence

Ordered Sequence:
1. Kafka Cluster Broker Health Check (All 5 brokers in ISR)
   ➔ 2. Consumer Group State Verification (State != Dead)
   ➔ 3. Replica Scale-out to match 16 partitions
   ➔ 4. Stream Processing Verification & Latency Stabilization

Explicit Stop/Escalate Gate: If consumer group stays in state 'PreparingRebalance' for > 3 minutes, STOP scaling. Escalate to Messaging Platform SME.

7. Edge Cases & Escalation Protocol

• **Edge Case:** Poison Pill Message in partition: Message with malformed schema crashing consumer continuously. Quarantine message using dead-letter-queue tool.

• **Edge Case:** Zookeeper / KRaft Metadata Quorum desync: Do not touch offsets.

Primary Contact: Kafka Platform Specialist (messaging-ops@fxtrading.internal | Slack: #kafka-prod-support)	**Secondary / Bridge:** Market Risk Technology Lead (risk-tech-oncall@fxtrading.internal)
Severity / SLA: P2 High: 20-minute resolution SLA	**Escalation Condition:** Total lag continuing to increase 15 minutes after scaling consumer instances.